



Case study

The computer science case study provides the stimulus to investigate a scenario involving current developments, emerging technologies and ethical issues in computer science.

The case study for SL is a scenario that includes two challenge questions that stimulate the required research. The information obtained will prepare students to answer the questions in this section of the examination.

HL students conduct deeper research into the case study which is reflected in the extra two challenge questions in the case study for HL, additional recommended teaching hours and time during paper one.

Adapted from the IBDP Computer science guide

The case study is assessed in section B of Paper 1, as shown in the table below. There is a mix of short-answer questions and one essay question.

The short-answer questions typically focus on your understanding of the main ideas of the case study, along with the terminology contained within. The command terms for these questions are words such as **define**, **identify**, **outline** and **describe**. The last page of the case study contains a list of terminology that is specific to the case study, over and above the terminology of the course, which the IB will expect you to be familiar with in your exam responses.

The essay question is where you demonstrate the depth of research-based understanding you have gained about the topics presented within the case study. The essay question is worth half the marks of section B of Paper 1. The marking criteria for the essay question are fixed and are provided later in this chapter.

■ Case study key information

	Standard Level students	Higher Level students
Recommended teaching time allocation	15 hours	30 hours
Proportion of Paper 1 exam	12 of 50 marks (24%)	24 of 80 marks (30%)
Marks for the essay question	6 marks	12 marks
"Challenges faced" research prompts that form the basis of the essay question	2	4
Approximate working time for the essay question in the examination	18 minutes	36 minutes

Responding to the case study

Here is a suggested approach to understanding and researching the issues surrounding the case study, and preparing for exam questions on it. Use this as a starting point and then modify it to your personal approach to learning.

■ Step 1: Understand the text of the case study

- Read the case study. Highlight and identify key points.
- Prepare definitions for the terminology list provided at the end of the case study.
- Quiz and test yourself and your peers on terminology definitions and introductory concepts.

■ Step 2: Understand the technology in the case study

The case study will involve field(s) of emerging technology not otherwise covered by the course syllabus.

- Identify resources such as video lectures and technical articles that provide an overview and introduction to the technologies present in the case study.
- Prepare notes based on those technologies from the resources you found. Ensure you have a good understanding of how the relevant technologies work. You will not have to include any programming code as part of a case study examination question, but you should have a good working technical understanding of the issues.

■ Step 3: Consider the scenario of the case study in its proper context

Now, with your renewed understanding of the technologies within the case study, give the complete scenario another careful read.

- What is the case study really about?
- Identify the big issues of the case study.
- What is interesting about the case study?
- What is confusing?

■ Step 4: Consider the challenges of the case study

For each of the challenges identified within the case study:

- Research any background information on the challenge so you have an appreciation of its relevance.
- Why is this a challenge in the scenario presented?
- What are the implications of not being able to meet this challenge?
- What are some potential solutions to the challenge?

It is recommended to produce a revision document that contains a summary of the issues and technology pertinent to each of the challenges.

■ Step 5: Consider the relevance of the challenges beyond the case study

Research and identify real-world examples of the challenges. Add these examples to your revision document from the previous step. This step is crucial as it will empower you to refer to other, real-life examples relevant to the case study when writing your exam responses, thereby helping you demonstrate meaningful research.

■ Step 6: Review for exam questions

- The essay question will be based on one or more of the challenges that are identified at the end of the case study document.
- Brainstorm potential examination questions that focus on the challenges.
- Write practice essays for each of the challenges faced. Prepare a bullet-point list of key points to convert into revision index cards in the lead-up to the examinations.
- Swap practice essays with a friend and provide feedback based on the essay question marking criteria provided below.

Suggested strategies for case-study research

- 1 Use generative AI to read the case study and have it prepare a summary of issues for each of the challenges. An example prompt might be: “Given the case study on the theme of [CASE STUDY THEME], prepare explanatory bullet points on [CHALLENGE FACED].” Then, perform a literature review to determine the accuracy of each of the assertions made by the AI. (Do not accept the responses of the AI at face value!)
- 2 Arrange with your classmates to present to or peer-teach each other about the various challenges, and then take questions from your class. Your confidence when answering those questions will help identify areas you need to research further.
- 3 Read major research papers and publications relevant to the case study so you can speak about developments in the field with authority. Refer to the papers that discuss those developments as you speak.

Cornell University runs the arXiv.org service, which is an excellent starting point: “arXiv is a free distribution service and an open-access archive for nearly 2.4 million scholarly articles in the fields of physics, mathematics, computer science, quantitative biology, quantitative finance, statistics, electrical engineering and systems science, and economics. Materials on this site are not peer-reviewed by arXiv” (arxiv.org).

As the chief examiner notes: “Students who read journal articles and complete video courses on the case study topic have a broader understanding of the concepts. This additional reading allows students to employ references to real-world examples or research in the extended response question” (*IBDP Computer science subject report*, May 2024).
- 4 Watch interviews or lectures given by technologists who are working with, or helped develop, the technology. For example, for the 2023 case study on recommendation systems, there are many excellent lectures available on YouTube by engineers from Netflix and Spotify, who discuss in great technical detail how their algorithms work.
- 5 Ask your teacher to arrange a class visit to a company or organization that is working with the technology. Meet with their team and interview them.

Consider, for example, the 2019 case study on computer-aided dispatch systems for emergency-service vehicles. The author arranged a class visit to a nearby app-development company that was creating an emergency-services app for the local government. This gave the students a firsthand account they could reference in their exam response.
- 6 Contact your local universities and ask to speak with professors or research thesis students who are conducting research in fields of Computer Science that are relevant to the case study. Most universities have PhD and Masters students conducting research in the fields of emerging technology that case studies tend to draw upon.

7 Find relevant connections with real-world usage of the technologies.

For example, for the 2024 case study on rescue robotics, several students discussed the use of rescue robots in the 9/11 disaster and compared it to the use of rescue robots to search buildings damaged in the Ukraine conflict. Students were able to evaluate how the technology had changed in the intervening 20 years.

● Top tips!

- Know what to expect before walking into the exam. Be sure you understand the structure of the paper, including the command terms and requirements of each question, particularly the extended response.
- Depending on the topic, see whether you and your classmates can organize a field trip or a relevant guest speaker. It would be great if you could refer to some primary research in your answers ("the time I visited the ...").
- Take five minutes in the exam to plan your extended response before you start writing. Any extended writing is vastly improved by spending just a few minutes planning.
- Ensure that any assertions you make are fully substantiated and underpinned by balanced analysis.
- Use appropriate Computer Science terminology throughout your response.
- Arrange with your classmates to present to or peer-teach each other about the various challenges, and then take questions from your classmates. Your confidence when answering those questions will help identify areas you need to research further.
- Use generative AI to read the case study and have it prepare a summary of issues for each of the challenges. Use the generated output to perform a literature review of the summary to determine the accuracy of assertions it makes. Due to the well-known and documented issues of hallucinations, do not rely on generative AI outputs without performing an independent review!



Common mistakes

- It is very easy to tell when a student doesn't know the terminology and is making it up. Incorrect use of terminology implies poor understanding of the issues, so will considerably limit the marks you can earn. Do not neglect to have a thorough understanding of the terminology list provided.
- If you are going to provide a citation in an examination response, don't make it up! As indicated, the examiners will be broadly familiar with the major papers and publications relevant to the case study.
- Many low-performing essays only regurgitate the content given in the case study itself. The exam marker already knows what is in the case-study document; they want to see what additional insights you can provide from your research beyond the case study.
- Another common error is not to provide a balanced analysis. For a top-band response, you must be nuanced enough to present multiple perspectives from which you draw a conclusion.
- Pay close attention to the wording of the questions and, in particular, their command terms. The type of response depends upon the command term, and merely writing down everything you know is not an effective strategy.
- "Several unofficial online forums have provided questionable information and guidance. You should research broadly and cross-check information found" (*IBDP Computer science subject report*, May 2024).

Essay question marking criteria

■ Essay question marking criteria for Standard Level students

Marks	Description
0	No knowledge or understanding of the relevant issues and concepts. No use of appropriate terminology.
1–2 Basic	Minimal knowledge and understanding of the relevant issues. Minimal use of appropriate terminology. The answer may be little more than a list. No reference is made to the case study or independent research.
3–4 Adequate	A descriptive response with limited knowledge and / or understanding. A limited use of appropriate terminology. There is some analysis. There is evidence of some research.
5–6 Competent	Knowledge and understanding of the related issues and / or concepts. Uses terminology appropriately in places. There is evidence of analysis. There is evidence of research. There is a conclusion.

SL Paper 1 markscheme, page 8

■ Essay question marking criteria for Higher Level students

Marks	Description
0	No knowledge or understanding of the relevant issues and concepts. No use of appropriate terminology.
1–3 Basic	Minimal knowledge and understanding of the relevant issues or concepts. Minimal use of appropriate terminology. The answer may be little more than a list. No reference is made to the information in the case study or independent research.
4–6 Adequate	A descriptive response with limited knowledge and / or understanding of the relevant issues or concepts. A limited use of appropriate terminology. There is limited evidence of analysis. There is evidence that limited research has been undertaken.
7–9 Competent	A response with knowledge and understanding of the related issues and / or concepts. A response that uses terminology appropriately in places. There is some evidence of analysis. There is evidence that research has been undertaken.
10–12 Proficient	A response with a detailed knowledge and clear understanding of computer science. A response that uses terminology appropriately throughout. There is competent and balanced analysis. Conclusions are drawn that are linked to the analysis. There is clear evidence that extensive research has been undertaken.

HL Paper 1 markscheme, pages 13–14

Analysing past case studies

This section provides a high-level overview of some recent case studies to illustrate the typical style in which they are presented and the essay questions that may appear as a result.

Additionally, chief examiner feedback on the quality of the essay responses is provided to help highlight common weaknesses in responses given by students to these questions. You will quickly notice common themes that echo throughout the examination seasons as you read those comments. Take the time to learn from the mistakes of those who have gone before you.

■ Blockchain (May & November 2020, May 2021)

- **Scenario:** Based on the mayor of Santa Monica, Pablo, who wants to establish a local currency, called MONS, that uses blockchain technologies.
- **Technologies:** Technologies referred to include:
 - ☐ digital ledgers
 - ☐ digital signatures
 - ☐ proof of work (mining)
 - ☐ structure of a blockchain
 - ☐ Merkle trees
 - ☐ SHA256.
- **Challenges:** There are a number of challenges that are linked to the introduction of MONS; these include:
 - ☐ understanding how new blocks are added to the ledger and how the proof of work prevents malicious nodes from taking over the MONS network

- understanding how the MONS architecture is scalable and can remain efficient as the number of users increases
- understanding the use of cryptographic techniques in the MONS project
- explaining to the Santa Monica citizens how their MONS balance is calculated from transaction data securely stored in a publicly accessible blockchain ledger
- investigating how the distributed nature of a blockchain cryptocurrency and the confirmation process may have disadvantages for the citizens of Santa Monica.
- **Questions:** Essay questions asked include:
 - Pablo states: “In a traditional banking system, users trust the banks to keep everyone’s money safe; but with MONS, the whole blockchain, right from the very first transaction, would be visible to all MONS users, so it is important to be able to explain to citizens how their money is guaranteed to be safe” (lines 109–112).
With reference to the key technologies, to what extent do you believe the MONS project will ensure the safety of the residents’ money? (November 2020)
 - Pablo has claimed that the use of blockchain technology for the MONS cryptocurrency will mean the cryptocurrency is both secure and scalable.
To what extent do you agree with Pablo? (May 2021)
- **Feedback:** The chief examiner, summarizing the quality of responses to this question, stated:
The majority of responses covered immutability, distributed consensus, cryptographic hashes, digital signatures, and the 51% attack. **Better candidates were able to reference real-world situations**, but most candidates did not. Candidates tended to identify increasing the blocksize and adjusting the nonce difficulty. Very few candidates had understood how sharding and layer 2 protocols could assist with scalability.
Many candidates discussed economic and environmental issues with blockchain, rather than the Computer Science aspects of security and scalability. **This approach lacks focus and detracts from the overall response.**

IBDP Computer science subject report, May 2021

■ Genetic algorithms (May & November 2022)

- **Scenario:** Based on the travelling salesperson problem.
- **Technologies:** Technologies referred to include:
 - genetic algorithms:
 - population
 - selection algorithms
 - crossover
 - mutation.
- **Challenges:** There are a number of challenges associated with genetic algorithms; these include:
 - understanding the role of convergence in genetic algorithms and the factors affecting convergence
 - evaluating the use and implementation of roulette-wheel selection, tournament selection and truncation selection strategies used within genetic algorithms
 - discussing the different solutions for addressing the failure of simple crossover strategies for the travelling salesperson problem; in particular:
 - why they are necessary
 - how they are applied

- how they preserve the parental traits
- what other possible methods are available
- understanding the advantages and disadvantages of genetic algorithms with respect to other approaches to the travelling salesperson problem, and to combinatorial optimization problems in general.
- **Question:** Essay questions asked include:
 - To what extent do the characteristics of genetic algorithms make them an appropriate approach to solving the route optimization problems? (May 2022)
- **Feedback:** The chief examiner, summarizing the quality of responses to this question, stated:

The challenge was to understand the advantages and disadvantages of genetic algorithms with respect to other approaches to the travelling salesperson problem and combinatorial optimization problems in general.

The broadness of the question meant that candidates could incorporate information from the other three challenges. Perhaps, for this reason, many candidates failed to demonstrate more than adequate understanding of how genetic algorithms can arrive at a successful solution and this approach's main weaknesses.

The majority of responses were descriptive rather than evaluative. Many candidates wrote incorrect statements or did not understand how the different characteristics affect each other. There were very few proficient responses, and **it was rare to see journal articles referenced.**

IBDP Computer science subject report, May 2022

- **Quotation:** The case study states:
 - Successful implementations of genetic algorithms strike a natural balance between exploration and exploitation, and techniques such as simulated annealing can fine-tune that balance as the algorithm progresses towards convergence (page 7, Discussion).
- **Question:**
 - Discuss the role of convergence in genetic algorithms and how exploration and exploitation can affect its success. (November 2022)
- **Feedback:** The chief examiner, summarizing the quality of responses to this question, stated:

Some candidates produced excellent answers demonstrating an understanding of how initial routes, population size, selection method, crossover method and mutation affect convergence. Proficient answers critically analysed the interplay between these choices and how they affected exploration and exploitation. Unlike previous case studies, there was no opportunity to reference real-world situations.

Many candidates only talked about exploration and exploitation in the broadest sense, failing to identify how they could be manipulated. Some students tried to involve details from earlier questions, stating the same content without a consistent structure.

Even though there was little opportunity to reference real-world scenarios in this case study, **some candidates referenced research papers** when they defined critical terms. **A reference to a relevant, newsworthy article or research paper is recommended for Question 4.** Schools that approach the case study from a research perspective provide context for students. For example, researching practical applications of genetic algorithms will increase students' conceptual understanding and make the topic more engaging.

IBDP Computer science subject report, November 2022

■ Recommendation systems (May & November 2023)

- **Scenario:** Based on a new application that allows users to view the work of artists who have yet to be discovered. Artists may include actors, singers, screenwriters, comedians, painters, sculptors and filmmakers. In fact, any artist who wants to demonstrate a talent will be able to upload files to the application. The uploaded content can be rated by all users. Based on their ratings, the application recommends new content to each user.
- **Technologies:** Technologies referred to include:
 - ☐ cloud computing
 - ☐ machine learning
 - ☐ recommender systems:
 - content-based filtering and collaborative filtering
 - k-nearest neighbours
 - matrix factorization
 - training of recommender systems
 - evaluating recommender systems.
- **Challenges:** To help with this new business venture, called NextStar, there are a number of challenges that you need to research:
 - ☐ understanding the similarities and differences between supervised learning, unsupervised learning and reinforcement learning
 - ☐ understanding how the k-NN algorithm and matrix factorization can be used within recommender systems
 - ☐ understanding how to train, test and evaluate a recommender system
 - ☐ comparing content-based filtering and collaborative filtering recommender systems
 - ☐ understanding the ethical concerns linked to the collection, storage and use of users' behavioural data.
- **Question:** Essay questions asked include:
 - ☐ “Recommender systems can use content-based filtering, collaborative filtering, or a combination of both. Hybrid recommender systems combine several machine learning algorithms.” (lines 43–44)
Discuss the advantages and disadvantages of these different approaches for building a recommendation system. (May 2023 TZ1)
- **Feedback:** The chief examiner, summarizing the quality of responses to this question, stated: Most candidates focused on content-based and collaborative filtering, **describing their process with little evaluation**. Some candidates stated challenges such as popularity bias, overfitting and cold start but failed to explain strategies to deal with these issues.

The majority of candidates wrote an adequate response. **Few referenced the case study**, and their analysis was on generic movie-recommendations systems. Often the examples were YouTube-related rather than NextStar. There were very few proficient responses, and it was rare to see journal articles referenced.

IBDP Computer science subject report, May 2023 TZ1
- **Question:**
 - ☐ Discuss whether the challenges associated with the development of an effective recommender system can be overcome through the choice of algorithm, training data and methods of evaluation. (May 2023 TZ2)

- **Feedback:** The chief examiner, summarizing the quality of responses to this question, stated: **The majority of candidates wrote an adequate response.** While many referenced NextStar, often their analysis was on a generic movie-recommender system.

The better responses honed in on the challenges and built their essay around them, offering ways to address them. These candidates understood that the type of recommender system might start off simple and evolve over time.

There were few proficient responses. **Most candidates failed to structure their essays** to focus on the challenges while explaining the computer science relating to recommender systems. **Very few candidates referenced journals** or other sources, and although some included real-world recommender systems, **it was rarely as a comparative analysis.**

The broadness of the question troubled some students who tried to include all three areas and **ended up with no depth** whatsoever. Their responses were usually **descriptive and lacked any analysis or evaluation.**

IBDP Computer science subject report, May 2023 TZ2

- **Question:**

- ☐ A recommender system that uses a supervised learning K-nearest neighbour (k-NN) algorithm is selected. Supervised learning algorithms require several decisions to be made, including setting any hyperparameters, choosing data sets, training and testing procedures, and evaluation strategies.

Discuss whether the selection and implementation of a k-NN algorithm in the development of a recommender system will give precise and accurate recommendations. (November 2023)

- **Feedback:** The chief examiner, summarizing the quality of responses to this question, stated:

Many students had not researched this topic and wrote basic responses, often copying several lines from the case study. Even those students who wrote adequate responses often failed to explore the algorithm beyond describing its operation. Technical terminology tended to be used appropriately but not expanded upon.

The better responses explored supervised learning, analysed the k-NN algorithms, and discussed how the recommender system could be evaluated. Weaknesses such as the cold start problem, popularity bias, and overfitting were considered. Very few responses considered strategies for overcoming these problems.

There were few proficient responses. However, those were analytical and considered the question from a broad perspective, even considering data sets and preprocessing strategies. **One or two students referenced journals or other sources.**

IBDP Computer science subject report, November 2023

■ Rescue robots (May & November 2024)

- **Scenario:** Based around rescue robots, which are designed to help with the search and rescue of humans after a disaster, such as an earthquake or tsunami. These robots may assist the efforts of rescue teams by searching and mapping areas, assessing damage, removing debris, delivering supplies and evacuating casualties.

- **Technologies:** Technologies referred to include:

- ☐ computer vision
- ☐ visual simultaneous localization and mapping (vSLAM)
- ☐ pose estimation.

- **Challenges:** In the development of the new rescue robot, the design team at BotPro face a number of challenges:
 - understanding how vSLAM navigates an environment with unknown obstacles and contours
 - minimizing the time rescue robots spend scanning and learning an environment
 - estimating the pose of people despite varying light and environmental conditions and body-part or multiple-object occlusion
 - updating existing maps in a dynamically changing environment, such as an earthquake where rubble is still shifting
 - developing an understanding of the ethical considerations of using autonomous robots in life-and-death situations.

■ **Question:** Essay questions asked include:

- A governmental department of disaster management is considering deploying rescue robots made by BotPro that use computer vision technologies for rescue operations in closed spaces, such as buildings and factories.

Discuss the benefits and costs of deploying rescue robots that use the vSLAM process and pose estimation techniques to carry out rescue operations in closed spaces.

(May 2024 TZ1)

- **Feedback:** The chief examiner, summarizing the quality of responses to this question, stated: The extended response challenge was to discuss the benefits and costs of deploying rescue robots that use the vSLAM process and pose estimation techniques to carry out rescue operations in closed spaces.

Most students **focused excessively on ethical points**, such as the safety of deploying robots rather than human rescue teams. When technical concepts were mentioned, this was done **without any depth**. Few students **referenced information beyond the case study**, giving the impression that little research had been conducted.

IBDP Computer science subject report, May 2024 TZ1

■ **Question:**

- vSLAM algorithms are designed to operate in GPS-denied or GPS-degraded environments. Rescue teams in these environments cannot rely on GPS tracking. BotPro wants your opinion on whether rescue robots installed with vSLAM algorithms will be effective when looking for injured or unconscious people in an emergency situation, such as a factory fire. Such emergencies are time critical. Evaluate the effectiveness of robots that use vSLAM algorithms to find casualties in an appropriate timeframe. (May 2024 TZ2)

- **Feedback:** The chief examiner, summarizing the quality of responses to this question, stated: The better responses provided a balanced account, displaying that the effectiveness of rescue robots can vary depending on several factors, including the environment, resources available, computational complexity, the quality of the sensor data and the complexity of the search area. Moreover, they demonstrated an awareness of factors such as battery life, availability of factory blueprints and operational time. **Proficient responses provided real-world examples of situations where rescue robots had been deployed and referenced research papers.**

Weaker responses either focused on the ethics of deploying rescue robots or merely listed the technologies without describing or evaluating them

IBDP Computer science subject report, May 2024 TZ2